Chapter 17

Lecture 17: Compactness. Locally compactness.

Definition 17.0.1 (def1). *X is compact if any open covering admits a finite subcovering.*

Definition 17.0.2 (def2: Frechet compactness). *X is limit point compact if every infinite subset of X has a limit point in X.*

Definition 17.0.3 (def3: sequential compactness/Bozano-Weierstrass compactness). *X is sequentially compact if every sequence $\{x_n\} \subseteq X$ has a convergent subsequence converging to a point in X.*

Theorem 17.0.1. *If X is a metric space, then X is compact if and only if it is limit point compact if and only if it is sequentially compact.*

Proof

The proof is quite long and technical. We will only show some directions in the following.

Theorem 17.0.2. *For an arbitrary topological space, compactness implies limit point compactness. The converse is not true in general.*

Proof

Let X be compact, and let $A \subseteq X$ be infinite.

We assume that A has no limit point. Then $A = \bar{A}$. So $X \setminus A$ is open. Furthermore, for each $a \in A$, we can choose a neighborhood U_a of a such that $U_a \cap (A \setminus \{a\}) = \emptyset$.

Then $\{U_a : a \in A\} \cup \{X \setminus A\}$ is an open covering of X. Since X is compact, there exists a finite subcovering, say $\{U_{a_1}, U_{a_2}, \cdots, U_{a_n}\} \cup \{X \setminus A\}$. Thus,

$$X = \bigcup_{i=1}^{n} U_{a_i} \cup (X \setminus A) = (X \setminus A) \cup \{a_1, a_2, \cdots, a_n\},$$

which implies that A is finite, a contradiction. Hence, A has a limit point in X.

■

Example 17.0.1 (Counterexample for the converse). *Let $Y = \{p, q\}$ with anti-dicrete topology $J_Y = \{\emptyset, Y\}$. Let $X = \mathbb{N} \times Y$ with the product topology where $\mathbb{N}$ has the discrete topology. Then every non-empty set $A \subseteq X$ has a limit point. Because if $(n, p) \in A$, then any open set containing (n, q) intersects A at (n, p). So X is limit point compact. However, X is not compact. Because $\{\{n\} \times Y : n \in \mathbb{N}\}$ is an open covering of X which admits no finite subcovering.*

Example 17.0.2 (Limit point compact but not sequentially compact). *Let X be defined as the above example. Consider the sequence $\{(n, p)\}_{n=1}^{\infty}$. This sequence has no convergent subsequence. Because for any point (m, p), the open set $\{m\} \times Y$ contains only finitely many terms of the sequence; for any point (m, q), the open set $\{m\} \times Y$ also contains only finitely many terms of the sequence. So X is not sequentially compact.*

Theorem 17.0.3. *For a first-countable topological space X, limit point compactness implies sequential compactness.*

Proof

Take a sequence $\{x_n\} \subseteq X$.

If the set of values $\{x_n : n \in \mathbb{N}\}$ is finite, then there exists a value x that appears infinitely many times in the sequence. So the subsequence constantly equal to x converges to x.

Suppose the set of values $\{x_n : n \in \mathbb{N}\}$ is infinite. Since X is first-countable, we can construct a countable basis $\{U_k\}$ at a limit point a such that

$$x_{n_1} \in U_1, x_{n_2} \in U_2, \cdots, x_{n_k} \in U_k, \cdots \tag{17.1}$$

and

$$U_1 \supseteq U_2 \supseteq \cdots \supseteq U_k \supseteq \cdots \tag{17.2}$$

That is for each open set $U \ni a$ there exists N such that $\forall k \geq N$, $U_k \subseteq U$. Then $x_{n_k} \in U_k \subseteq U$ for all $k \geq N$. So $x_{n_k} \to a$. ■

Theorem 17.0.4. *Sequential compactness implies compactness for metric spaces.*

Proof This is harder than other directions.
Exercise Prove that X is sequentially compact, then it is limit point compact.
Remark For topological spaces, we have the following facts:

1. Compactness ⇒ Limit Point Compactness
2. Sequential Compactness ⇒ Limit Point Compactness
3. But other implications are not true in general.

17.1 Locally Compact

Definition 17.1.1. *A space X is said to be locally compact at x if there is some compact subspace C of X containing a neighborhood U of x. If X is locally compact at each of its points, then X is said to be locally compact.*

Example 17.1.1. $\mathbb{R}$ *is locally compact. Because for any $x \in \mathbb{R}$, take $U = (x-1, x+1)$ and $C = [x-1, x+1]$ which is compact.*

Example 17.1.2. $\mathbb{R}^n$ *is locally compact.*

Example 17.1.3 (non example)**.** $\mathbb{Q} \subseteq \mathbb{R}$ *is not locally compact.*

Exercise Show the above example.
Solution Let C be a compact subspace of $\mathbb{Q}$ containing a neighborhood U of $x \in \mathbb{Q}$. Since U is open in $\mathbb{Q}$, there exists an interval $(a, b) \subseteq \mathbb{R}$ such that $U = (a, b) \cap \mathbb{Q}$. We know $\mathcal{Q}$ is dense in $\mathbb{R}$, so there exists an irrational number $y \in (a, b)$. Let $\{q_n\} \subseteq (a, b) \cap \mathbb{Q}$ be a sequence converging to y. Since C is compact, so $\{q_n\}$ has a subsequence converging to a point in C. But a convergent sequence in $\mathbb{R}$ has a unique limit, so the subsequence converges to y, which is not in $\mathbb{Q}$ thus not in C. This is a contradiction. Hence, $\mathbb{Q}$ is not locally compact. ■

Example 17.1.4 (non example)**.** $\mathbb{R}^\omega$ *with product topology is not locally compact. If U is open, then U contains a basis element $(a_1, b_1) \times \cdots \times (a_n, b_n) \times \mathbb{R} \times \mathbb{R} \times \cdots$. Then U is not contained in any compact set C (C contains one factor of $\mathbb{R}$, then the projection to some factor space is $\mathbb{R}$, but a projection is continuous, which means $\mathbb{R}$ is compact. This is definitely not true).*